

# Hrishi Kabra

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## Education

### Tulane University | School of Science and Engineering

Present — May 2028

*Bachelor of Science, Major: Computer Science and Mathematics*

*New Orleans, LA*

- Minor: Artificial Intelligence and Business | GPA: 3.95/4.0 | Dean's List – Fall 2024, Spring 2025, Fall 2025
- South Asian Scholar (merit-based full tuition award - \$67,000 per year), First Year Honors Scholar
- Relevant Coursework: Algorithms, Data Science, Linear Algebra, Multivariate Calculus, AI Engineering

## Experience

### Niyogin AI

June 2025 — July 2025

*AI Intern*

*Mumbai, India*

- Developed and shipped a production-grade, end-to-end NLP pipeline using custom SpaCy named entity recognition (NER) to reliably extract structured information from noisy OCR output across financial and identity documents
- Achieved ~97% extraction precision and automated KYC and form digitization workflows to reduce manual review
- Engineered a Python-based document processing workflow to convert diverse multi-format and scanned bank statements into structured, query-ready Excel reports, streamlining downstream data ingestion and analysis

### Tulane University Department of Computer Science

August 2025 — Present

*Teaching Assistant*

*New Orleans, LA*

- Spring '26 – CMPS3160: Introduction to Data Science; Fall '25 – CMPS1500: Introduction to Computer Science I
- Facilitate weekly lab sessions and office hours for 50+ students, delivering hands-on instruction in Python programming, data analysis, pandas, and core computer science concepts to diverse student cohorts
- Collaborate with faculty to review course materials, design assignments, and provide structured feedback to reinforce and deepen student understanding of analytical and programming fundamentals

### Tulane Data and Artificial Intelligence

July 2025 — Present

*Research Assistant*

*New Orleans, LA*

- Co-authored paper accepted at AAMAS 2026 (Conference on Autonomous Agents and Multiagent Systems)
- Designed and optimized positional scoring models using simulated annealing, gradient descent, and mixed-integer programming (MIP) to optimize decision outcomes over large-scale preference datasets
- Contributed to the optimal\_voting Python package and Django web interface, enabling researchers to find and compare interpretable, optimal positional scoring rules across four social welfare functions

## Projects

### F1 Rule & Penalty Interpreter | Python, Gradio, LLM APIs, RAG

February 2026 — Present

- Engineered an end-to-end LLM-powered system over 1000+ FIA regulation and steward decision PDFs, enabling retrieval and generation of rule-grounded explanations for Formula 1 incidents through a RAG-based pipeline
- Designed evaluation pipelines and conducted GenAI Ops experiments on chunking, embedding retrieval, and context size, improving citation coverage from 60% to 93% while analyzing trade-offs between recall, noise, and reliability

### Reefguardian | Python, AWS (S3/Glue/Athena)

September 2025 — Present

- Designing a serverless AWS data pipeline to ingest and process NOAA global Sea Surface Temperature (SST) datasets at scale, enabling real-time visualization of coral bleaching hotspots for marine researchers
- Building automated ETL workflows using AWS Glue and Athena to support structured querying and low-latency retrieval of large-scale oceanographic datasets, minimizing manual data intervention and errors

### FishID | Python, Flask, Next.js, TypeScript, SQLite | ([fishid.vercel.app](https://fishid.vercel.app))

June 2025 — Present

- Built and deployed a full-stack web application that classifies fish species from user-uploaded images, integrating Fishial and Groq AI APIs through a RESTful Flask backend with a persistent fish identification log
- Implemented JWT and bcrypt authentication with split deployment across Vercel and Render for persistent storage

## Skills and Interests

**Languages:** Python, Java, C, C++, JavaScript, TypeScript, SQL, R, Assembly, HTML/CSS, Bash

**Technologies:** ClauReact, Next.js, Flask, Django, PyTorch, Pandas, AWS (S3, Glue, Athena), Docker, SQLite, Git

**Concepts:** Data Structures & Algorithms, REST APIs, Prompt Engineering, NLP, RAG, LLM Agents, Optimization, MLOps, ETL & Data Pipelines, Full-Stack Development, Systems Programming, Cloud Data Engineering, Fine Tuning

**Spoken Languages:** English (Fluent), Hindi (Fluent), German (Conversational)

**Certifications:** PADI Master Scuba Diver, PADI Rescue Diver, Emergency First Responder

**Interests:** Scuba Diving, Underwater Videography, Formula 1, Wordle, Monopoly, Strategy Games